

they swamped the Germans with their sheer weight of numbers. France had grouped together aerial divisions of 700 bombers and fighters. By the time of the St. Mihiel offensive in September, US General William "Billy" Mitchell commanded a force of 1,500 French, British, and American aircraft in his sector of the front. Exhausted, short of fuel and spares, and with their airfields increasingly exposed to air attack, the German airmen never gave up the fight and, indeed, inflicted heavier losses than they suffered. But it was a struggle driven by despair rather than hope.

World War I ended with the armistice of November 11, 1918; the Germans surrendered without truly accepting defeat. The war had cost an estimated nine million lives. Of those, probably some 15,000 were airmen. This may seem a relatively small figure in absolute terms, but individually a pilot probably stood no better chance of surviving the war than an infantryman in the trenches. Aviation had come of age in a war of mass slaughter driven by industrial technology. Although many people saw airmen soaring above that impersonal butchery, they had fully played their part as victims and killers. Flight had lost its innocence.

FALLEN WARRIOR

Allied soldiers examine the wreckage of a German aircraft after one of the last battles of the war. Although the German pilots were often better trained than their opponents and had at least marginally superior aircraft, they could not cope with the number of Allied machines filling the skies.

BOMBS FROM THE AIR

DROPPING EXPLOSIVES ON PEOPLE on the ground was one of the first conceived uses for aircraft. During World War I this primitive urge was refined into strategic bombing (such as factories and cities); tactical bombing of targets behind the front line (including railroads or supply dumps); and frontline attack. The earliest bombs were artillery shells dropped over the side of aircraft by hand.

By 1917, sturdier, specialized bomber aircraft had appeared with bomb racks, bomb sights, and release systems. Raids were carried out by day when air superiority allowed it, or by night with much reduced accuracy. By the final stages of the war, aircraft roamed the front, bombing bridges and airfields, and strafing troops and trucks, proving beyond a doubt the vital importance of air superiority.



FLECHETTE

Feathered flight helps guide 5-in (12-cm) dart

MARTEN HALE BOMB

Shell contains 4½ lb (2kg) of explosive

Propeller guides bomb

INCENDIARY BOMB

Perforated "carcass" casing helps bomb catch fire on impact

BOMBS AND ARROWS

The bombs of both sides at the start of the war were small and basic. The British also used flechettes – dartlike weapons released in boxes of 500 from around 5,000ft (1,500m) – but they were ineffective against infantry and cavalry concentrations.

NIGHT BOMBARDMENT

This painting shows a night bombardment by a Voisin biplane. Night operations made bombing less accurate, but in daylight, bombers were easy targets for enemy fighters.

